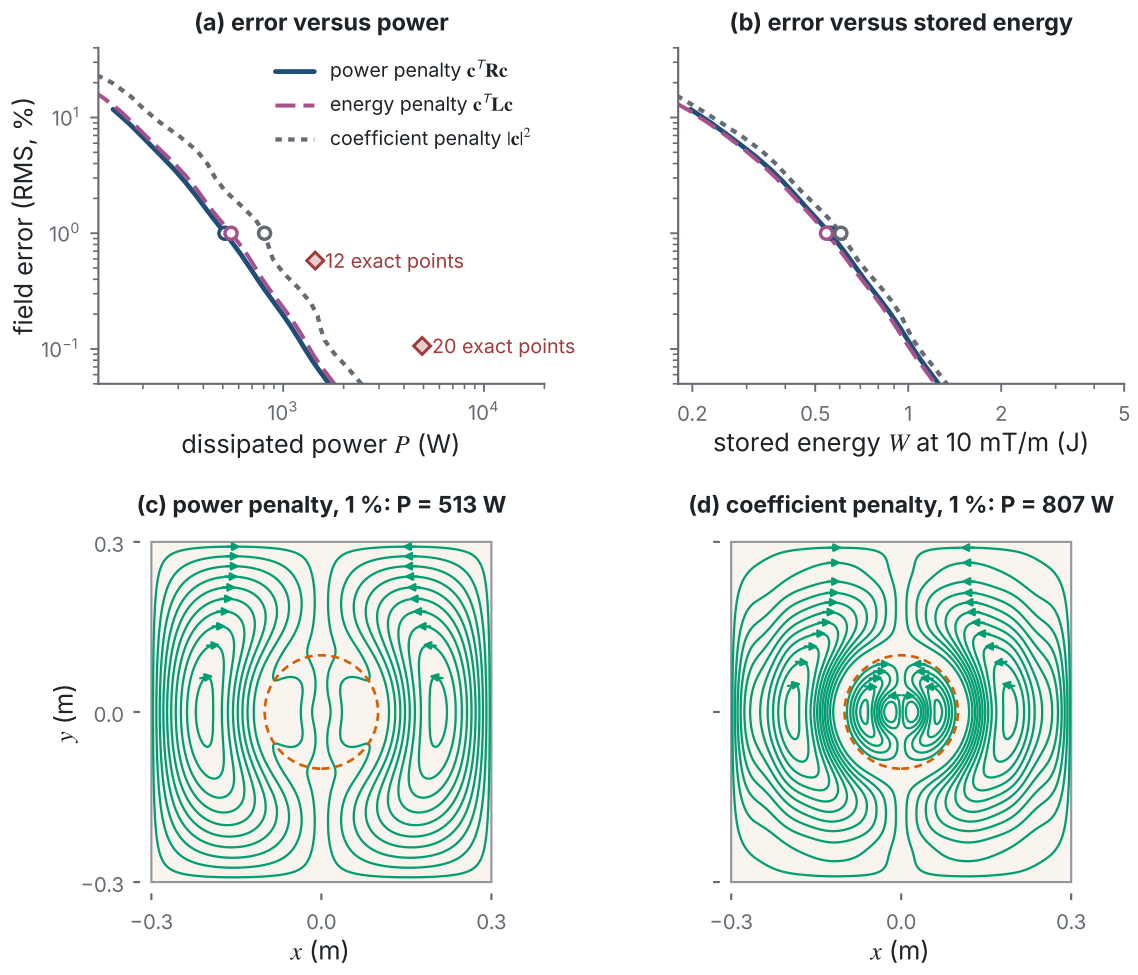

(a) error versus power
power penalty $\mathbf{c}^T\mathbf{R}\mathbf{c}$
energy penalty $\mathbf{c}^T\mathbf{L}\mathbf{c}$
coefficient penalty $|\mathbf{c}|^2$
12 exact points
20 exact points
field error (RMS, %)
10^1
10^0
10^{-1}
10^3
10^4
dissipated power P (W)
(b) error versus stored energy
0.2
0.5
1
2
5
stored energy W at 10 mT/m (J)
(c) power penalty, 1 %: P = 513 W
0.3
0.0
−0.3
y (m)
−0.3
0.0
0.3
x (m)
(d) coefficient penalty, 1 %: P = 807 W
−0.3
0.0
0.3
x (m)